$$\mathbf{P}\psi(x) = -i\hbar\frac{d\psi(x)}{dx}. \qquad (8.15)$$

Quantum physicists often use units in which $\hbar$ is exactly one, and in that way simplify the equations. As tempting as it is, we won't do that here.

Let's work out the eigenvectors and eigenvalues of **P**. The eigen-equation in abstract vector notation is

$$\mathbf{P}|\Psi\rangle = p|\Psi\rangle, \qquad (8.16)$$

where the symbol *p* is an eigenvalue of **P**. Eq. 8.16 can also be expressed in terms of wave functions. Using the identification

$$\mathbf{P} = -i\hbar\frac{d}{dx},$$

we can write the eigen-equation as

$$-i\hbar\frac{d\psi(x)}{dx} = p\psi(x)$$

or

$$\frac{d\psi(x)}{dx} = \frac{ip}{\hbar}\psi(x).$$

This is a type of equation that we've run into before. The solution has the form of an exponential:

$$\psi_p(x) = Ae^{\frac{ipx}{\hbar}}.$$